Step 2: Data Description

The automated pet feeder relies on several inputs to function correctly. The real-time clock tracks the current time to trigger scheduled feedings. The food-level sensor monitors whether the food hopper has enough supply, and the bowl weight sensor measures the amount of food eaten by the pet.

The system produces outputs based on these inputs. The servo motor rotates to dispense food portions, while alerts notify the user if the food is uneaten, the hopper is empty, or a mechanical issue occurs.

The following table summarizes the inputs, outputs, sample values, and operational constraints of the system:

Data Table

Inputs/Outputs	Name	Descriptions	Sample values	Operational Constraints
Input	Real-time clock	Tracks current time for feeding schedule	08:00, 18:00	Pre-programmed & stores up to 7 feeding times
Input	Food-level sensor	Detects whether food storage is empty/full	Full, Empty	Binary reading only (no partial measurement)
Input	Weight Sensor	Measures food weight in grams	0g–500g	Accuracy ±2g & alert if change <5g after 10 mins
Output	Servo Motor	Rotates to dispense food	0°–180° rotation	Requires 12V DC & fixed portion rotation time
Output	Alerts (LED/SMS & Buzzer)	Notifies user of events or gives audible warning	"Food low", "Uneaten food", On/Off	Needs network/power & only triggers for critical alerts & no alert history stored